

Seismic Loading Framework Tutorial

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1. Consider two buildings, A and B , both with design lives of 50 years. Building A is a generic office building and should be designed for a level of ground-motion that has a 10% chance of being exceeded over its life. Building B is a critical facility for which the consequence of failure is very high. The probability of exceedance for the design level ground-motion for Building B should only be 1%.

(a) Compute the ratio of the return periods for buildings B and A .

(b) If the average annual rates of exceedance of ground-motion levels im is defined by:

$$\lambda(IM > im) = 0.05 \exp(-2im)$$

compute the corresponding ratio of design ground-motion levels for the two buildings.

2. Imagine that the vulnerability of a building can be expressed in terms of the way that the probability of failure, P_f , varies with resistance, r . Further imagine that a particular relationship between these quantities is:

$$P_f(r) = \exp(-r/2)$$

and assume that the cost of failure is 1000 monetary units.

A given building system, $S1$, can provide a level of resistance according to a cost function:

$$C_{C,S1}(r) = 100 + 5 \exp(2r)$$

(a) Determine the optimal level of resistance for this building and the associated total cost.

(b) Another building system, $S2$, can provide the same level of resistance for cheaper:

$$C_{C,S2}(r) = 100 + 5 \exp(1.5r)$$

What is the reduction in total cost associated with using system $S2$ over $S1$?

- (c) What is the change in the probability of failure if this alternative system $S2$ is adopted over system $S1$?